BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022 • Paper Code: BN-IX-XXII-I

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A (Version-C)

Time Allowed: 20 Min | **Maximum Marks:** 12 | **Paper Code:** BN-IX-XXII-I

Note: There are THREE sections in this paper i.e. A, B & C. Attempt section A and return it to the superintendent within the given time. You have four choices for each objective type question as A, B, C and D. The choice which you think is correct fill that circle in front of the question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question.

Q1. The relation between coefficient of linear and volume expansion is.

(A) $\alpha = 3\gamma$
(B) $\gamma = 3\alpha$
(C) $\gamma = \alpha/2$
(D) $\gamma = 6\alpha$

Q2. In gases, heat is mainly transferred by

(A) Conduction
(B) Convection
(C) Radiation
(D) Both A & B

Q3. A measuring cylinder is used to measure

(A) Mass
(B) Area
(C) Volume
(D) Level of liquid

Q4. A car starts from rest. It acquires a speed of $25ms^{-1}$ after 20s. The distance moved by the car during this time is

(A) 31.25 m
(B) 250 m
(C) 500 m
(D) 5000 m

Q5. A body moving with uniform velocity. What will be its acceleration?

(A) Uniform
(B) Zero
(C) 10 ms^{-2}
(D) -10 ms^{-2}

Q6. Force needed to produce an acceleration of 10 ms^{-2} in a ball of mass 0.5 Kg is

(A) 20 N
(B) 10.5 N
(C) 9.5 N
(D) 5 N

Q7. The weight of body is 147 N, its mass will be _____ when g = 9.8 ms^{-2}.

(A) 14.7 N
(B) 1.47 N
(C) 15 N
(D) 15.7 N

Q8. The unit of torque is ;

(A) Nm^{-2}
(B) Ncm
(C) Nm^{-1}
(D) Nm

Q9. The orbital speed of geostationary communication satellite is _____.

(A) 3.07 Kms^{-1}
(B) 3.07 Kmh^{-1}
(C) 30.7 Kms^{-1}
(D) 30.7 Kmh^{-1}

Q10. The radius of the earth r_e is

(A) 9.8 m
(B) 6.67×10^{-11} m
(C) 6×10^{24} m
(D) 6.4×10^{6} m

Q11. A 2 Kg hammer head moving with a speed of 4 ms^{-1} has a kinetic energy of _____.

(A) 32 J
(B) 16 J
(C) 8 J
(D) 4 J

Q12. The unit of strain is

(A) Kgm^{-3}
(B) Pa
(C) Nm^{-2}
(D) None

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 32 | **Paper Code:** BN-IX-XXII-I

> Note: Attempt section B & C accordingly.

Q2. (Marks 32) Q.No 2. Attempt any EIGHT parts of the following. All parts carry equal marks.

i. How least count for Vernier caliper and screw gauge are defined?

ii. Write the number given below in scientific notation. a. 3000000 ms^{-1} b. 6400000 m c. 0.0000016 g d. 0.000054 g

iii. A cyclist moving with uniform acceleration of 1.2 ms^{-2}. How much time will it require to change his velocity from 6 ms^{-1} to 12 ms^{-1}.

iv. Define coefficient of kinetic friction. Write advantages and disadvantages of friction.

v. Why it is better to use a long spinner rather than a short one to loosen a rusty nut?

vi. Gravitational force is an inverse square force. Explain

vii. A person pulls a suitcase through the airport at 45° angle. The tension in the rope is 20N. How much work does the tension do if the suitcase is pulled 100m?

viii. Define power. What is its unit? Calculate the relation between kilowatt hour and Joul.

ix. Which material is more elastic, steel or rubber and why?

x. The temperature of a normal human body is 37°C. Find this temperature on the Fahrenheit and Kelvin scale.

xi. Why is ice at 0°C a better coolant of soft drinks than water at 0°C?

SECTION – C

Maximum Marks: 21 | **Paper Code:** BN-IX-XXII-I

Note: Attempt any THREE of the following. All questions carry equal marks

Q3.

a) Prove third equation of motion graphically i.e $2as = vf^2 - vi^2$

b) With what speed must a ball be thrown vertically from ground level to rise to a maximum height of 50m?

Q4.

a) Derive an acceleration and tension in Atwood's machine.

b) A bolt on a car engine needs to be tightened with a torque of 40 Nm. You use a 25 cm long wrench and pull on the end of the wrench perpendicularly. How much force do you have to exert?

Q5.

a) State and explain the law of universal gravitation.

b) The maximum height a typical human can jump is about 60 cm. By how much does the gravitational energy increase for a 72 Kg person in such a jump?

Q6.

a) State and explain Pascal's principle.

b) How much heat is required to increase the temperature of 0.5 Kg of water (with specific heat capacity of 4190 $JKg^{-1}K^{-1}$) from 10°C to 65°C?